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Adversarial-resilient lightweight phishing url detection: Evaluating lexical & metadata features under evasion techniques

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Phishing is considered one of the most widespread and dynamic cyber threats, as attackers use deceitful URL structures to circumvent traditional detection tools. Despite promising performances of machine learning-based phishing detection techniques, the majority of existing models are evaluated under clean conditions, failing to account for adversarial URL evasion methods including obfuscation, encoding manipulation, homoglyph substitutions, token padding, and subdomain reordering. This weakness grossly compromises their practical performance in real-world deployments. Phishing URL detection is modelled as a lightweight binary classification problem supported by an explicit adversarial threat model. Lexical, structural, and *metadata-based* URL features are represented comprehensively without relying on raw URL strings or deep packet inspection (DPI), preserving user privacy and enabling broad deployment. To mitigate adversarial vulnerabilities, the Adversarial-Resilient Lightweight Random Forest (AR-LRF) model is proposed, combining controlled ensemble complexity with simulated adversarial perturbations applied during training. Extensive experiments on a large-scale imbalanced real-world dataset (91% benign, 9% phishing) consisting of 650,000 URLs were evaluated across accuracy, precision, recall, F1-score, and ROC-AUC under clean and adversarial conditions against five baseline classifiers: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Naïve Bayes. The AR-LRF achieves 99.78% accuracy and ROC-AUC of 0.9999 on clean data, with significantly lower performance degradation under adversarial perturbations. Feature importance and confidence stability analyses further validate the model's resilience, positioning AR-LRF as a viable, lightweight, and deployable tool for modern cybersecurity infrastructure.

Keywords Phishing URL detection, Adversarial machine learning, Cybersecurity, Lightweight classification, Feature engineering, Random forest, URL evasion attacks

The increased pace of digitalization of the contemporary society has dramatically changed how people, organizations, and governments relate to the Internet. Thousands of new domains are registered every day and millions of web requests are launched, and this results in a greater threat surface that cybercriminals can use. Malicious Uniform Resource Locater (URLs) have become one of the most successful tools in undertaking cyberattacks including phishing, malware spread and defacing websites in this so-called hyper-connected environment[1]. These illogical URLs usually present themselves as valid web addresses defrauding users into disclosing vital information, executing malicious codes, or accessing confidential systems without authorization. As a result, the identification of malicious URLs has become a staple of cybersecurity studies and a precondition of protection of personal and organizational properties. In general, cybersecurity can be described as a set of measures aimed to protect computer systems, networks, and other digital infrastructures against agendas that aim to undermine confidentiality, integrity, or availability[2]. Out of all the variety of cyberthreats, the most common is phishing because it is both psychologically and technically easy. Phishing attacks exploit users by sending them fraudulent links in emails, in social-media messages, or spoofed sites to steal credentials or execute malware programs. A worldwide warning-intelligence study has shown that phishing alone cost billions of dollars annually and is ever-increasing in magnitude and complexity[3]. One of the most threatening figures published by RSA (2013) was that over 450 000 websites had been attacked through phishing activities in one year and that the cost of this attack amounted to USD 5.9 billion[4]. These figures highlight the dire need to have

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credible and dynamic detection systems that can detect malicious URLs that have never been seen before before they are defined.

Blacklists and rule-based filters are viable protection mechanisms that continue to be deployed widely due to their simplicity and low cost of computation. Nevertheless, these systems are reactive in nature, i.e. they are only able to block URLs that are already known to be malicious[5]. Such lists become outdated very fast because of the constant dynamism of cybercriminals; attackers can conveniently bypass fixed filters by doing minimum textual change like character replacement, inserting random tokens and using short-link obfuscating services[6]. This cat-and-mouse game has caused research focus to be on intelligent detection systems that have the capability of generalizing beyond threats that have been explicitly known. Machine learning (ML) thus has turned into a central method of characterizing the behavioral and lexical properties of malicious web links[7]. ML-based detection systems make classification of links as either benign or malicious based on discriminative features extracted using URL strings, domain properties or metadata. Trained classification algorithms like Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) have been shown to be good at training nonlinear decision surfaces that reflect finer details between legitimate and malicious URLs[8]. Random Forest models are superior in large-scale studies with regard to accuracy and robustness because they are ensemble-based in averaging features and toleration to noise[9]. Classical ML detectors despite their prediction success have their flaws in dealing with the changing adversarial manipulations. As the models are trained on fixed datasets, any small structural perturbations even homoglyph replacement like replacing g00gle by google or subdomain reordering can severely decrease the accuracy[10].

New developments in phishing detection have investigated deep learning and hybrid models to overcome these constraints. Multi-layer feature representations and behavioral analysis are found in systems like PhiUSIIL (Computers & Security), and StealthPhisher (Expert Systems with Applications) to help them detect as well as under advanced attacks. Moreover, attention-based CNN-LSTM models have also shown high potential in sequential and contextual dependency on URLs, which can be better utilized to detect obfuscated phishing links. Although they are highly accurate, these methods tend to have heavy computational costs, which restricts their use in real time and resource-constrained settings.

To address these constraints, recent research emphasizes lightweight and adversarially robust architectures. Such models are particularly important for deployment in constrained environments such as mobile devices, email gateways, and Internet-of-Things (IoT) nodes, where computational and memory resources are limited[11]. A recent work in privacy-aware UAV communication, secure energy sharing, and blockchain supported cyber-physical systems further underscores the critical need for lightweight and reliable, distributed security architectures in resource-limited environments[40–42]. These works do not have a specific focus on phishing detection, but they contribute to the broader requirement of deployable security models that maintain efficiency, security and operational reliability in edge and IoT environments. Rather than enforcing strict deployment constraints such as sub-50 KB model size, this study focuses on maintaining low computational overhead while ensuring real-time inference capability. At the same time, adversarial training methods have been proposed to improve model resilience by exposing classifiers to synthetically generated adversarial variations during training[12]. These approaches aim to transform conventional classifiers into adaptive detectors capable of handling evolving phishing strategies. Explainable artificial intelligence (XAI) is another emerging area in cybersecurity. Techniques such as SHAP and LIME enable interpretation of model predictions by highlighting the influence of specific features[13]. This enhances user trust and supports analysts in understanding model behavior and refining detection rules.

The current study is situated within this evolving research landscape and proposes an Adversarial-Resilient Lightweight URL Detection Framework. Moreover, privacy-preserving and collaborative federated or mobile-cluster learning frameworks are a recent development that highlight the emerging importance of edge-centric and collaborative intelligence in dynamic networks[43,44]. The directions are consistent with the goal of the present work, to build a phishing detector that is computationally feasible yet robust against adversaries. The framework integrates lexical and metadata features with adversarial training and explainability mechanisms to achieve robust detection performance under realistic conditions. Unlike prior methods that focus primarily on efficiency[14], the proposed approach emphasizes resilience to adversarial perturbations while maintaining interpretability and scalability.

The general objective of this work is to develop and evaluate a phishing detection model that maintains high precision and low false positives under adversarial conditions. The specific objectives are as follows:

1. To design a lightweight machine learning model using lexical and metadata features for efficient phishing URL detection.
2. To model realistic evasion techniques such as homoglyph substitution, token padding, subdomain manipulation, and encoding-based obfuscation.
3. To incorporate adversarial training strategies and evaluate robustness using metrics such as F1-score and Robust-AUC.
4. To evaluate model performance on a large-scale aggregated dataset reflecting real-world class imbalance.
5. To integrate explainability mechanisms for interpreting model predictions.

The proposed research contributes across multiple dimensions. Scientifically, it formalizes an adversarial threat model for URL classification, an area that remains underexplored in lightweight detection systems. Technologically, it presents a practical solution for deployment in constrained environments by balancing detection performance and computational efficiency. From an engineering perspective, the study provides a reproducible evaluation framework for adversarial robustness. Socially, it contributes to safer web usage by reducing false alarms and mitigating economic losses caused by phishing attacks[15,16]. The combined focus on resilience, transparency,

and efficiency distinguishes this work from existing approaches that rely on static datasets and aligns with the growing demand for deployable and trustworthy AI-based cybersecurity solutions. More recent energy-efficient low-rank tuning and secure cyber-marketplace studies also suggest that the prospective systems should be aware of the deployment cost, the preservation of privacy, and the resource-aware operation along with predictive performance[45–47]. This is in line with the practicalness of the proposed lightweight AR-LRF framework.

The rest of this paper is organized as follows. Section 2 presents the literature review covering existing detection mechanisms, feature engineering strategies, and adversarial challenges. Section 3 describes the proposed methodology, including dataset construction, feature extraction, model design, and adversarial training. Section 4 discusses experimental results, including detection performance and robustness analysis. Finally, Sect. 5 concludes the study and outlines directions for future research.

Literature review

Malicious URL-based and phishing attacks remain one of the most severe and dynamic areas of threat to the cybersecurity sector. Phishing is not like conventional malware because it uses deception and social engineering, which contributes to its high adaptability and challenge to counter with the help of conventional defenses. The initial countermeasures were more of blacklist-based and rule-based filters but they have shortcomings of slow response, weak generalization and low tolerance to new phishing URLs and obfuscated phishing URLs. Therefore, in the last decade, the emphasis of research has moved to the active, learning-oriented detection mechanisms that can detect previously unseen attacks based on the structural, lexical, and behavioral patterns that are encoded within the URLs and web resources[17–19].

Recent research points out that effective phishing URL detection is a multi-dimensional challenge that needs to take special care in combining feature engineering, lightweight but expressive models, adversarial manipulation resistance, and interpretability to enable trust in analysts and operational use[16,18,20]. This part summarizes the literature available at four interdependent dimensions: (i) feature modalities and engineering strategies, (ii) classical machine-learning methods and efficiency issues, (iii) deep and hybrid learning architectures, and (iv) adversarial robustness and explainability. The review ends with the determination of research gaps that remain open to support the current research.

Feature modalities and feature-engineering strategies

The basis of the URL-based phishing detection is still in feature representation. The earliest and most popular methods are based on lexical and syntactic characteristics directly derived out of the URLs, e.g. length of URLs, amount of digits and special characters, entropy, token frequency, use of suspicious keyword (e.g., login, verify, secure), and the use of IP addresses rather than domain names[21–24]. Such handcoded interfaces are computationally cheap and ideally applicable in real-time detection of latency-needy applications, such as email gateways and browser extensions.

In order to supplement lexical cues, metadata-based features have been proposed, to obtain contextual information on top of surface structure. These are the top-level domain (TLD) category, the use of HTTPS, and the age of the domain, the details of the registrar, the number of redirection, and the ranking of web popularity [25,[26,27,21]. Combining lexical and metadata is proven many times by empirical studies to be a more effective method of enhancing detection performance, as it encodes both superficial and contextual cues of phishing[16,21,24].

Since there are limitations to what can be deployed, some works explore feature selection and dimensionality reduction methods to still have small models without losing accuracy. Variance thresholding, minimum redundancy maximum relevance (mRMR), entropy-based selection, and information-theoretic filtering are some of the methods that have been used to minimize redundancy and model complexity[21,28,29]. Although lightweight lexical-metadata feature vectors can be highly-performing on curated datasets, they are susceptible to targeted obfuscation, and hybrid and robustness-oriented methods are investigated in the subsequent sections.

Classical machine learning: strengths, efficiency, and limits

The classical supervised learning algorithms continue to be popular in phishing URL detection, as they are easy to interpret, can be trained quickly, and come with low inference costs. Logistic Regression, Decision Trees, Random Forests, Support Vector Machines and k-Nearest Neighbors have shown high baseline results on various datasets[21,30,31,24]. Of these, Random Forests will always be superior to other classifiers because of their ensemble characteristic, ability to resist noise and affinity of exploiting interaction between features that are nonlinear[21,34].

The significance of efficiency-oriented learning is mentioned in a number of studies, especially when it comes to large-scale URL corpora. Decision tree based on features and cost-sensitive learning models have been suggested to minimize false positives and still be scalable[30]. Streaming models Detection frameworks like PhishStorm also testify to the fact that classical models can be scaled to real time high-throughput scenarios[27]. Classical models have significant weaknesses despite these advantages. To begin with, they implicitly rely on stationary data distributions, which are very vulnerable to evasion techniques, including homoglyph replacement, token padding, and subdomain manipulation[32,33]. Second, they do not have the ability to model sequential or hierarchical patterns of raw URL strings. Such inadequacies have prompted the use of deep and hybrid learning architectures.

Deep learning and hybrid architectures

Phishing URL detection has been receiving more and more interest based on deep learning techniques because they can learn hierarchical and sequential representations. Convoluted neural networks at the character level, recurrent neural networks, and CNN-RNN models have been used to learn the finer details of obfuscation and contextual factors in URLs[34,35,24,36]. Contextual embedding approaches that use transformers, such as BERT-

style models, have achieved the current state of the art in detection accuracy through the modeling of long-range dependencies and semantic relationships[37].

New phishing detection systems also have come up with new architectures that incorporate both a hybrid and a security-profile approach. The similarity-index and incremental-learning model for phishing URL detection was proposed in Computers & Security, whereas StealthPhisher proposed a hybrid deep-learning and GenAI-based phishing defense framework in Expert Systems with Applications[40,41]. In these studies, the emphasis is shifting toward the use of more features, adaptive learning and better representation strategies to detect phishing. Nevertheless, deep models are expensive in terms of computational and memory, hindering their use in edge and real-time application settings. In this regard, recent research paper discusses hybrid models that combine handcrafted lexical and metadata with lightweight neural embeddings and strike a balance between expressiveness and efficiency[16,38]. Optimization and pruning methods have been used to enhance accuracy as well as restrain the model complexity through evolutionary methods[39].

Even though hybrid and deep models are better than classical models in clean environment, their resilience to adversarial perturbation and their interpretability is a critical issue, especially to operational cybersecurity systems.

Adversarial behavior and robustness

Evidence manipulation of URLs constitutes one of the biggest problems in phishing. Malicious actors actively use the vulnerabilities in the trained feature distribution through homoglyph replacement, Unicode encoding, URL shortening, over padding, and swapping subdomain/TLD[32,33]. It is experimentally observed that models that are only trained on clean datasets fail to perform significantly worse on adversarial perturbations, frequently leading to a significant loss of F1-score and recall[32].

Two major approaches have been developed to deal with this weakness. The former is adversarial data augmentation/adversarial training where realistic evasion instances are introduced to the training procedure to prompt invariant decision boundaries[32,39]. The second is concerned with powerful feature engineering, including measures based on entropy, structural sobriety, and metadata indications that are more immune to exterior perturbations[16,26]. Along with that, continuous and life-long learning systems have also been suggested to keep the detection models up to date with the changing phishing tactics with time[39]. Though they look promising, these methods come with issues of acquisition of labels, catastrophic forgetting, and complexity of deployment.

Explainability and trust analyst

In addition to the accuracy of detection, interpretability has emerged as a key issue to real-world phishing detection systems. SHAP and feature attribution methods, as explainable AI (xAI), have been used to explain why a URL is considered malicious by pointing out the powerful lexical and metadata properties[20]. Explainability makes analysts more confident in their work, levels regulatory compliance, and provides debugging of users of the model through the exposure of brittle or over-dominant features.

Recent research has shown that smaller and understandable models tend to be intuitive to humans, detecting some suspicious signs of embedded IP addresses, unnatural TLDs, or excessive encoding[20]. Although it is becoming increasingly popular, explainability is yet to be included in adversarial robust phishing detection pipelines in a systematic way.

Recent state-of-the-art models (2022–2025)

The recent developments of phishing URL detectors have more and more resorted to deep learning and hybrid architectures to address the drawbacks of the classical machine learning methods. Transformer-based models, especially those based on Bidirectional Encoder Representations in Transformers (BERT) have shown good performance in terms of identifying contextual and semantic relationships within URLs. These models encode URLs as sequences and learn the complicated dependencies among tokens, allowing it to effectively identify more sophisticated obfuscation patterns. Nevertheless, their complexity makes them costly to compute and memory intensive, restricting their usefulness in real time and resource limited situations.

Hybrid models such as Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) have also been popular with phishing detection. CNN elements work well in extracting local variations like suspicious tokens and character-level features whereas LSTM networks extract sequential dependencies on the URL structure. This combination enables CNN-LSTM models to detect not only structural anomalies but also temporal patterns, which leads to a higher detection accuracy than single models. Recently, there have been attention-based architectures that are proposed to improve interpretability and performance. Attention mechanisms allow the model to put emphasis on the most pertinent components of the URL, e.g., abnormal subdomains, encoded strings, or suspicious keywords, and enhance the accuracy of the classification and the explainability of the results. Hybrid attention-based CNN-LSTM models also combine feature learning and attention weighting with state-of-the-art results in a number of studies.

Although effective, these deep learning methods usually have a high computational cost and are not efficient to deploy. This weakness drives the desire to develop lightweight but strong models that can be able to strike a balance in performance in terms of detection, adversarial resilience, and real-time use.

Comparative synthesis and open research gaps

Taken collectively, the literature suggests that the use of lexical and metadata attributes with classical machine-learning models produce high baseline-performance and deployment-efficiency[21,24]. Deep and hybrid are an additional layer of protection against detection and obfuscation, but come at the cost of computational complexity and interpretability[37,39,34]. The adversarial training enhances robustness, but available assessment is narrow in

depth, variety of data, and verisimilitude of attacks, and the comparability of cross-studies is challenging[28,40]. Elucidable detection methods are under-represented in deployable systems.

Table 1 is a summary of representative research, which compares the feature modalities, model type, datasets and limitations. The classical machine-learning models are efficient, interpretable, and do not have the strength to adversarial evasion. Deep and hybrid models are more accurate and compromised by a high computational cost and low transparency. There are adversarially aware and explainable models being developed, but few papers consider lightweight deployment, adversarial robustness, and interpretability as one unified and reproducible framework.

Research methodology

Problem formulation and threat model

The problem of the phishing URL detection was developed into a binary classification problem where the task was to identify a benign or phishing URL based on the given URL. The feature vectors were feature vectors of the observable lexical, structural and metadata features of each URL. Let $x \in R^d$ denote the extracted feature vector for a URL, and $y \in \{0,1\}$ represent the corresponding class label, where 0 indicates a benign URL and 1 denotes a phishing URL. The detection task can be mathematically expressed as:

$$f(x) \rightarrow y, y \in \{0,1\} \tag{1}$$

where $f(\cdot)$ represents the learned classification function.

To model the decision confidence of the classifier, the posterior probability of a URL being phishing given the feature vector x was defined as:

$$P(y=1|x) = \sigma(f(x)) = \frac{1}{1+e^{-f(x)}} \tag{1a}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, mapping the classifier output to a probability score in the range $[1]$.

Besides the mainstream classification environment, the present study directly factored in an adversarial evasion threat model, which is a realistic phishing environment. The opponent was supposed to have the ability to be able to modify URLs without changing its ill motive. Common evasion techniques were URL obfuscation by just padding the tokens, added misleading subdomains, and manipulations based on encoding including hexadecimal or Unicode character substitution. These methods are used to provide distortion in feature distributions and maintain functional redirection behavior.

Unlike prior formulations that rely solely on feature-level perturbations, this study also considers realistic string-level adversarial transformations applied directly to raw URLs before feature extraction.

On the side of the defender, the detection system was limited to work on a lightweight deployment platform, which only used URL-based features. Webpage rendering, deep packet inspection and external content fetching were deliberately omitted with the aim of keeping the computational overhead low and it should operate in real-time. Figure 1 is used to depict the interaction between the evasion mechanisms of the attacker and the detection pipeline of the defender.

Realistic URL-level adversarial transformations

To more closely match the adversarial model to real-world phishing tactics, adversarial perturbations are used on the raw URL strings prior to feature extraction. Simulated attacker behavior is used to simulate these transformations, which in turn affect the generated feature distributions.

Adversarial samples were created in a two-step process for the revised experimental design. First, the raw URL strings were directly transformed to a realistic level using URL transformations. Second, the transformed

Ref.	Study (Year)	Feature Types Used	Detection Model / Approach	Key Contribution	Major Limitation
16	Prabakaran et al. (2023)	Lexical + Structural + Metadata	Deep learning with feature fusion	Demonstrated accuracy gains via multi-feature integration	Computationally intensive; no adversarial evaluation
17	Kavya et al. (2024)	Multiple (surveyed)	Systematic review	Comprehensive synthesis of modern phishing detection trends	No empirical validation
18	Asif et al. (2023)	URL lexical features	Survey (ML-based phishing detection)	Identified dataset bias and generalization issues	Lacked robustness and deployment analysis
20	Wang et al. (2025)	Lexical URL features	Explainable ML (SHAP-based)	Integrated explainability into phishing detection	Adversarial robustness not addressed
31	Yuan et al. (2024)	Lexical perturbations	Adversarial ML benchmarking (SpacePhish)	Exposed severe vulnerability of ML detectors under evasion	Did not propose lightweight defense
36	Elsadig et al. (2022)	Raw URL embeddings	BERT-based deep learning	Achieved high detection accuracy using contextual embeddings	High latency and memory footprint
37	Nagy et al. (2023)	Lexical + Sequential	Parallel & sequential ML pipelines	Improved detection through pipeline combination	Increased system complexity
39	Ejaz et al. (2023)	Lexical (streaming URLs)	Continual / lifelong learning	Addressed concept drift and evolving phishing attacks	Requires continuous labeled data

Table 1. Comparative Summary of top representative studies in phishing URL detection.

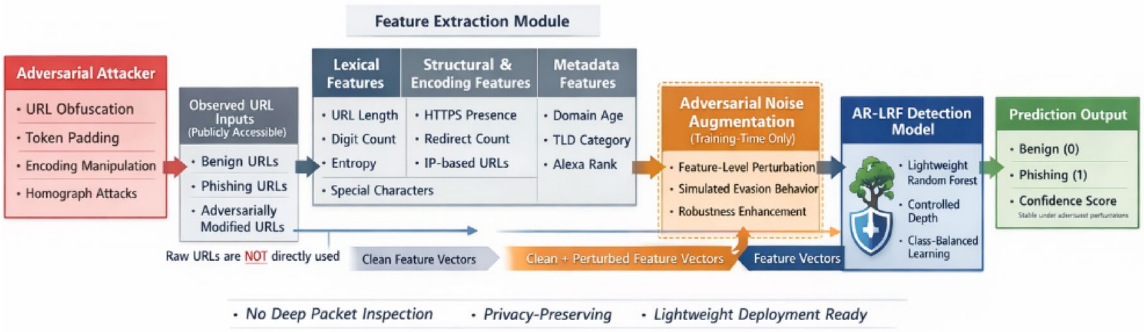


Fig. 1. Overall, threat model and phishing URL detection pipeline.

URLs were also processed again using the same feature extraction pipeline for obtaining new lexical, structural, encoding based and metadata feature vectors. In this regard, the samples generated for the main robustness test were not just mathematical noise on the feature vectors but were created through the manipulation of URL strings and then through feature extraction.

The adversarial transformations are those that are regarded as the following:

- Homoglyph substitution:

• *Example* • "google.com" → "g00gle.com" (character replacement using visually similar symbols).

- Token padding:

• *Example* • "/login" → "/login-secure-update" (insertion of misleading tokens).

- Subdomain manipulation:

• *Example* • "secure.bank.com" → "bank.secure-login.com" (reordering and insertion of deceptive subdomains).

- Encoding-based obfuscation:

• *Example* • "/login" → "%2Flogin" or Unicode-based transformations.

Once these transformations are used, the modified URLs are reprocessed to derive new lexical, structural and metadata features. This helps to make the adversarial effects appear in the feature space realistically as opposed to the synthetic noise that approximates the adversarial effects.

The adversarial-resilient model also features a joint decision-risk criterion to trade-off classification accuracy, robustness, and computational efficiency:

$$\mathrm{R}(\mathrm{x}) = \lambda_1 \mathrm{L}_{\mathrm{cls}}(\mathrm{x}, \mathrm{y}) + \lambda_2 \mathrm{L}_{adv}\left(\mathrm{x}, x'\right) + \lambda_3 \mathrm{C}(\mathrm{x})$$

where $\mathrm{L}_{\mathrm{cls}}(\mathrm{x}, \mathrm{y})$ represents classification loss, $\mathrm{L}_{\mathrm{adv}}(\mathrm{x}, \mathrm{x}')$ denotes adversarial consistency loss, and $\mathrm{C}(\mathrm{x})$ captures computational cost. The coefficients $\lambda_1, \lambda_2, \lambda_3$ control the trade-off between accuracy, robustness, and efficiency.

This formulation guarantees that the model is optimized not only in terms of predictive accuracy but is also robust to adversarial perturbations, as well as lightweight computational demands. Both string-level transformations and feature-level consistency are also included which makes the proposed threat model more realistic and robust.

Dataset description and ethical considerations

The experimental evaluation was conducted on a huge set of URLs with 650,000 instances to simulate realistic phishing pattern observed in web traffic. The acquired data was imbalanced with 91% benign and 9% phishing URLs, whereas artificially balanced data was nearly always a mix of 50–50. This distribution was deliberately kept to be reflective of the real-life scenario in which malicious web addresses account for a small proportion of all web traffic.

The data set was collected from open repositories commonly used for phishing detection research. Phishing URLs were gathered from OpenPhish and PhishTank, while benign ones were taken from Alexa and Tranco domain archives. The data was collected in 2024 to ensure variations in URL structures and attack patterns. To remove URLs that are inactive, duplicate, or malformed, they were duplicated and inactive, leaving 650,000 unique URLs after preprocessing. The URLs were converted into sets of numerical attributes based on lexical features (URL length, entropy, number of digits, number of special characters), structural features (number of subdomains, length of path, and complexity of query), and metadata features (domain age and popularity

rank). Raw URLs were not used directly for the input of the models, but were kept for feature extraction and auditing only. This means that the learning was directed towards general phishing attributes instead of learning the domain names or service providers.

The data set was split into training set (80%) and test set (20%) by stratified sampling, maintaining the same proportion of the classes. Moreover, a 5-fold cross-validation protocol was used to boost the robustness and repeatability of the reported results. No personally identifiable information (PII) was present on any of the URLs utilized in this study and the URLs were all publicly accessible. They did not gather any user-generated content, credentials or private communications. The approach helped to mitigate privacy concerns and align with ethical standards in cybersecurity research by limiting data used for learning to derived numerical attributes. These key characteristics of the data set are summarized in Table 2.

Feature engineering and representation

The feature engineering was very important in learning discriminating patterns between benign and phishing URLs. In accordance with the latest phishing detection research, the features were divided into three categories; lexical, structural and encoding-based features and metadata features. This clear division allowed controlled experimentation and made it possible to perform a robustness analysis.

The selected features were chosen based on their ability to capture both surface-level anomalies and deeper structural inconsistencies commonly exploited in phishing attacks. In particular, these features are lightweight, interpretable, and computationally efficient, making them suitable for real-time detection scenarios.

Lexical features

Lexical features were obtained directly out of the URL text and were intended to reflect superficial attributes that are usually related to phishing attempts. They were the total length of URL, the amount of numeric characters, the amount of special characters and the Shannon entropy. Phishing URLs tend to have an abnormal length and great randomness as a result of the obfuscation strategies.

For example, URL length is an important indicator of obfuscation, as attackers often insert additional tokens or parameters to disguise malicious intent. Similarly, entropy captures the randomness of character distribution, where higher entropy values are typically associated with encoded or artificially generated phishing URLs.

$$H = -\sum_{i} p_i log p_i \tag{2}$$

where p_i denotes the probability of occurrence of the i-th character in the URL string.

In order to ensure scale consistency across lexical features, entropy values were normalized as:

$$H_{norm} = \frac{H - H_{min}}{H_{max} - H_{min}} \tag{2a}$$

where H_{min} and H_{max} represent the minimum and maximum entropy values observed in the training dataset.

Structural and encoding features

There were structural features in which a URL was constructed and coded. These characteristics were: use of IP addresses rather than domain name, the use of HTTPS, number of redirections, and signs of hexadecimal or Unicode encoding. These features are particularly useful in identifying deceptive URL constructions, where attackers attempt to mimic legitimate domains using encoding tricks or misleading structural patterns. For instance, the use of IP addresses instead of domain names and excessive redirections are often indicative of malicious intent.

Metadata features

Metadata properties were used to record contextual data that was not part of the URL itself, such as age of domain, Alexa rank, and top-level domain (TLD). These features provide an additional layer of trust-based signals, where domain age serves as an indicator of credibility (newly created domains are more likely to be malicious), and popularity rankings such as Alexa rank help distinguish well-established websites from suspicious or low-traffic

Attribute	Description
Total instances	6,50,000
Benign URLs	591,500 (91%)
Phishing URLs	58,500 (9%)
Class distribution	Imbalanced (realistic)
Feature categories	Lexical, Structural, Metadata
Raw URL usage	Feature derivation only
Model input type	Numerical features only
PII involvement	None

Table 2. Summary of dataset characteristics.

domains. These features are much more prone to adversarial manipulation and thus selective inclusion was a feature of these features based on the experimental setting. Table 3 contains a detailed taxonomy of all features, their types, rationale and evasion vulnerability.

Baseline detection models

To set a standard and balanced standard for performance measurement, several popular machine learning classifiers were chosen as baseline models. They were Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Support Vector Machine with Radial Basis Function kernel (SVM-RBF), and Gaussian Naïve Bayes (GNB). These models are a mix of linear, nonlinear, probabilistic and ensemble learning, which are used in phishing URL detection research.

The updated evaluation protocol extends the baseline models used in the initial experimental setup to include a new baseline that explicitly focuses on fairness. In detail, a Random Forest-Full + Adv (RF-Full + Adv) model was trained with the same feature and adversarial augmentation strategy as the proposed AR-LRF framework. The baseline is based on the full feature representation (lexical, structural, metadata), as well as adversarially augmented samples generated under the same perturbation settings as described in Sect. 3.5.2. The addition of RF-Full + Adv has two points of importance. First, it removes the possibility of bias that can be introduced by the fact that the set of features used in the baseline models may differ from the set of features used in the proposed framework. Second, it allows to test the contribution of richer feature representations and adversarial augmentation to the improvements in performance, and the contribution of the architectural design choices that are part of the proposed adversarial-resilient training framework. As a result, the modified experimental setup can better allow for a comparison between conventional machine learning methods and the proposed method to be made, and in a statistically sound manner.

The original baseline models (LR, DT, RF, SVM and GNB) still run with low capacity lexical and structural features, corresponding to the phishing detection systems used commonly in practice reported in the literature and used for the detection of phishing attacks. Conversely, RF-Full + Adv is a strong, adversarially trained benchmark that is trained with the same feature and augmentation settings as AR-LRF. All of the experimental setups are listed in Table 4.

Proposed adversarial-resilient model (AR-LRF)

The essence of the paper is that there is a proposed framework Adversarial-Resilient Lightweight Random Forest (AR-LRF) which is designed to ensure a stable performance of phishing detection even in cases when URLs are deliberately altered using evasion methods. In contrast to the case of baseline classifiers that are only trained on clean feature distributions, AR-LRF explicitly uses adversarially perturbed samples in training to increase robustness. The proposed framework was made to be computationally lightweight to be deployed in real-time, and enhanced resilience to adversarial URL transformations. The objective of AR-LRF is to minimize classification error while maintaining robustness under adversarial perturbations and preserving computational efficiency. To ensure controlled and reproducible adversarial training, the perturbation magnitude and robustness constraints are explicitly defined. Adversarial perturbation strength is controlled by a fixed parameter, or $\delta = 0.1$ which defines the range of acceptable variation in the value of features. Also, a strong constraint threshold $\tau = 0.5$ is added to restrain too much distortion between original and perturbed feature representations.

These parameters guarantee that the produced adversarial samples are realistic and semantically sound, and still introduce a significant amount of variability to increase the resilience of the model to evasion attacks.

Model architecture

AR-LRF was constructed based on a lightweight Random Forest backbone, which was chosen because it performs well, is inexpensive to run inference, and resilient to noisy feature distributions unlike a purely linear model. The architecture used class-balanced learning to maintain constant decision boundaries between phishing and benign classes especially when using perturbed distributions. The tree depth was kept to a bound on model

Feature	Category	Description	Attack Sensitivity
URL length	Lexical	Total characters in URL	Medium
Digit count	Lexical	Number of numeric characters	Medium
Special characters	Lexical	Count of special symbols	Medium
Entropy	Lexical	Character randomness	Medium
HTTPS presence	Structural	Secure protocol indicator	Low
IP-based URL	Structural	IP instead of domain name	Low
Redirect count	Structural	Number of redirections	Medium
Hex encoding	Encoding	Hexadecimal characters	High
Unicode usage	Encoding	Unicode obfuscation	High
Domain age	Metadata	Age of domain (days)	High
Alexa rank	Metadata	Website popularity	High
TLD	Metadata	Top-level domain	Medium

Table 3. Feature taxonomy and attack sensitivity analysis.

Model	Key Hyperparameters	Feature Scope	Adversarial Augmentation
Logistic Regression	Max iterations = 2000	Lexical + Structural	No
Decision Tree	Max depth = 8	Lexical + Structural	No
Random Forest (RF)	200 trees, max depth = 12	Lexical + Structural	No
SVM (RBF)	RBF kernel, probability enabled	Lexical + Structural	No
Naïve Bayes	Gaussian assumption	Lexical only	No
RF-Full	200 trees, max depth = 12	Lexical + Structural + Metadata	No
RF-Full + Adv	200 trees, max depth = 12	Lexical + Structural + Metadata	Yes
Proposed AR-LRF	300 trees, max depth = 16	Lexical + Structural + Metadata	Yes

Table 4. Baseline models and experimental configuration.

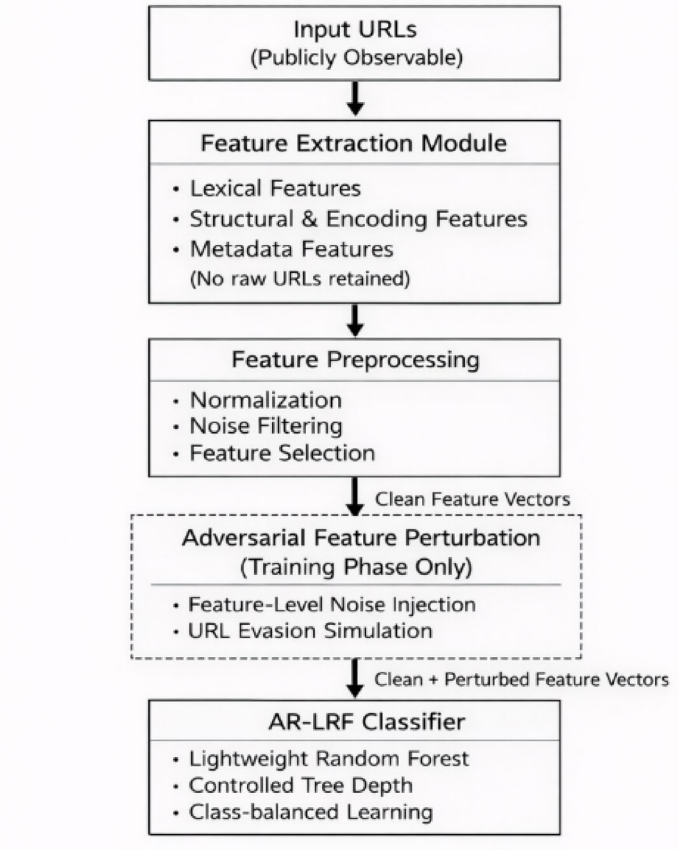


Fig. 2. Proposed AR-LRF architecture for adversarial-resilient phishing URL detection.

complexity, and a constraint on the minimum leaf size was also used that ensured that the model did not over-specialise to training patterns, and minimised overfitting.

Figure 2 shows the architectural design of AR-LRF, which consists of feature extraction, preprocessing, adversarial augmentation and final ensemble inference. This design explains why baselines (trained on clean samples only) vary with the proposed model (trained on clean + perturbed samples).

Adversarial noise augmentation

AR-LRF applied perturbations at the feature level perturbations to training in order to simulate real-world behavior of URL evasion. The revised implementation did not limit adversarial augmentation from being purely at the feature level of mathematical noise. The main adversarial training samples involved homoglyph substitution, padding the tokens, transforming the subdomain and encoding-based obfuscation of the raw URLs. These modifications were then run through the feature extraction stage again such that the adversarial feature vectors produced capture the measurable lexicostructural changes induced by such realistic URL manipulation.

Such a strategy is indicative of the fact that attackers can obfuscate URLs with token padding, manipulation of encode and structure parameters and structural modification, which shifts the values of lexical and structural features but does not alter the malicious motive. In this research, adversarial samples were created with the

help of controlled perturbations to specific features (e.g., length, the number of digits, the number of special characters, and entropy), and adversarial variants of the semantically plausible sample were created, which change the statistical representation in the sample.

In addition to adversarial augmentation using the URL, a bounded feature-level perturbation mechanism was employed for controlled sensitivity analysis. The study was able to explore the effect of model performance as the perturbation size δ is increased, without altering the other experiments that formed the main robustness experiment, which were based on transformed URLs.

Let x represent the original feature vector. The adversarially perturbed feature vector x' was generated as:

$$x' = x \cdot (1+)\,, \sim U\,(-\delta, \delta) \tag{3}$$

where δ controls the magnitude of perturbation and $U(\cdot)$ denotes a uniform distribution.

In this case, x is the feature vector after URL preprocessing and feature extraction. To obtain bounded perturbation variants for the sensitivity analysis, we employed URL-level transformed samples for robustness evaluation and principal adversarial training, while using Eq. (3) for URL-level variants.

This formulation had the advantage that feature perturbation was constrained in both space and value, avoiding large distortions which could not be realistic changes to URLs. To prevent unrealistic adversarial distortion, the perturbed feature vector was constrained as:

$$\| x' - x \|_2 \leq \tau \tag{3a}$$

where τ is a predefined robustness threshold ensuring that adversarial samples remain semantically valid and practically realizable.

Robust training strategy

AR-LRF training strategy was aimed at enhancing the generalization in case of adversarial conditions. In particular, the model was trained using a mixed set of clean and adversarially produced samples, so that the classifier will learn the decision boundaries that will be maintained even in the case the distribution of features changes as a result of evasion. This distribution-aware learning minimises overconfidence and enhances any consistency of predicted probabilities which is vital in testing phishing detection systems in adversarial manipulation.

The originality of such a strategy is that high percentage of lightweight phishing detection pipelines described in previous literature only train and test on clean URL distributions. The given method enhances the reliability of the detection in the situation when the attacker intentionally tries to mislead machine learning, which is achieved by explicitly including adversarially perturbed samples.

The proposed perception-decision pipeline shows the overall flow of the AR-LRF framework in Fig. 3. Raw URLs are converted first into structured feature representations and then adversarial augmentation is applied to model evasion attacks. The refined data is then fed into the lightweight ensemble model to make final classification decisions to make sure that it is not only robust but also real-time.

Evaluation protocol and experimental setup

Each experiment was carried out in accordance with a standard evaluation protocol to ensure fairness and reproducibility. Instead of relying solely on a single stratified train–test split, a 5-fold cross-validation strategy was adopted. In this setup, the dataset was divided into five equal subsets, where in each iteration, four subsets were used for training and one subset was used for testing. This process was repeated five times, and the final performance was computed as the mean ± standard deviation across all folds. A fixed random seed was maintained to ensure consistency and reproducibility of the splits.

Prior to dataset partitioning, duplicate URLs and/or duplicate domain level records were eliminated in order to reduce risk of data leakage. In addition to this, the learning algorithms did not receive raw URL strings, but only numerical features that were derived from the URL strings. The data set was divided into training and testing sets by stratified sampling technique so as to maintain the class distribution of the data set in both sets. These changes were made with the aim of minimizing the effects of memorization and to make the results reported more valid.

The F1-score was computed as:

$$F1 = \frac{2PR}{P+R} \tag{4}$$

where P denotes precision and R denotes recall. The robustness loss under adversarial conditions was quantified using accuracy degradation as:

$$\Delta_{robust} = \frac{Acc_{clean} - Acc_{adv}}{Acc_{clean}} \tag{4a}$$

where Acc_{clean} and Acc_{adv} denote classification accuracy under clean and adversarial conditions, respectively.

Additional security-oriented metrics

Further measures of reliability of the detection system in security-critical settings were taken through other error-specific measures. False Negative rate (FNR) and False Positive rate (FPR) were determined as:

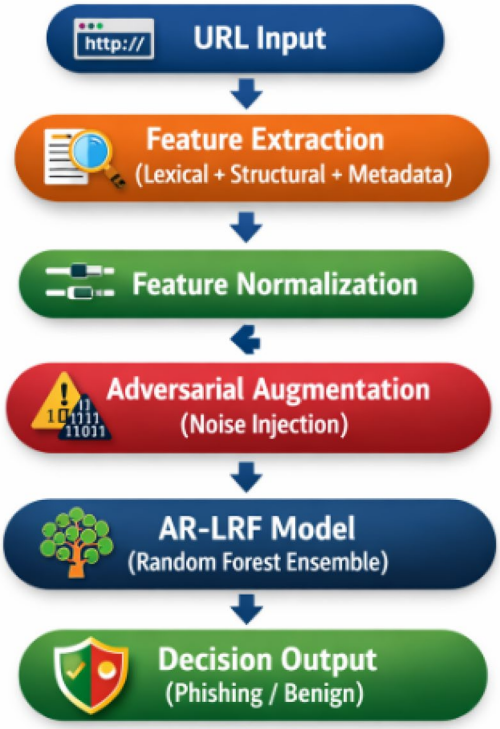


Fig. 3. Perception–Decision Pipeline for Adversarial URL Detection.

Model	Accuracy (Mean ± Std)
Logistic Regression	*0.968 ± 0.004*
Decision Tree	*0.984 ± 0.003*
Random Forest	*0.989 ± 0.002*
SVM (RBF)	*0.979 ± 0.003*
Naïve Bayes	*0.935 ± 0.005*
Proposed AR-LRF	*0.997 ± 0.001*

Table 5. Cross-validation performance summary (mean ± standard deviation).

$$FPR = \frac{FP}{FP + TN} FNR = \frac{FN}{FN + TP}$$

In which FP, TN, FN, TP are false positives, true negatives, false negatives, and true positives respectively. FPR represents the number of benign URLs falsely designated as phishing (false alarms), whereas FNR represents the number of phishing URLs falsely designated as benign (missed detections). These metrics are especially valuable in the field of cybersecurity where a false alarm and the failure to detect an attack also come with important operational consequences. The overall stability and consistency of the evaluated models under the cross-validation protocol are summarized in Table 5, where performance is reported as mean ± standard deviation across all folds.

Adversarial robustness assessment

The review of adversarial robustness is an essential necessity of the current phishing detection study, especially when the models are implemented in natural settings where the attacker is constantly evolving. Two conditions were used to evaluate robustness in this study: (i) the clean URLs condition and (ii) the adversarially perturbed URLs condition created with evasion perturbation mechanism, as defined in Eq. (3). This allowed one to directly compare the way features distributions are manipulated in terms of the ascent of baseline models and the proposed AR-LRF.

Metric	Definition	Interpretation
Accuracy	((TP + TN)/(TP + TN+FP + FN))	Overall correctness
Precision (P)	(TP/(TP + FP))	Reliability of phishing prediction
Recall (R)	(TP/(TP + FN))	Ability to detect phishing cases
F1-score	(2PR/(P + R))	Balance of precision and recall
ROC-AUC	Area under ROC curve	Ranking quality across thresholds

Table 6. Evaluation metrics and definitions.

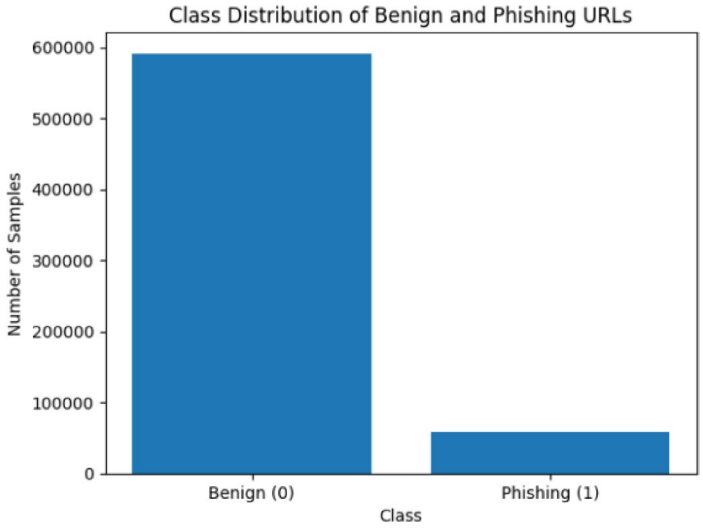


Fig. 4. Accuracy vs. perturbation magnitude (δ).

Implementation details and reproducibility

All experiments were implemented using standard machine learning libraries to ensure transparency and reproducibility. The implementation was executed in Python, leveraging widely adopted frameworks for preprocessing, model training, and evaluation. Random seeds were fixed at the preprocessing and train–test split stages to guarantee replicable data partitioning and consistent baseline comparisons. Hyperparameters for baseline models were selected using controlled configurations consistent with lightweight phishing detection literature, while AR-LRF parameters were tuned to balance robustness and computational efficiency without over-parameterization.

To enable reproducibility, the implementation explicitly documented: (i) feature subsets used for baseline and proposed models, (ii) perturbation bounds δ for adversarial augmentation, (iii) training set construction (clean + adversarial samples), and (iv) consistent evaluation metrics and reporting procedures. These measures collectively ensure that the experimental findings can be independently verified and extended in future work.

Sensitivity analysis

A sensitivity analysis was also performed to assess the strength of the proposed AR-LRF model to different degrees of adversarial perturbation by systematically changing the perturbation magnitude parameter δ. This evaluation offers insights into the performance of the model with respect to the increased adversarial strength and the stability of the suggested framework is confirmed. The perturbation parameter δ was manipulated between [0.05, 0.20] with incidences and both clean and adversarial performance was measured. Two aspects were examined (i) classification performance at different perturbation levels, and (ii) robustness decay compared to the clean performance benchmark.

In order to evaluate sensitivity four perturbation magnitudes were taken: δ = 0.05, 0.10, 0.15, and 0.20. In each perturbation level, adversarial samples were created by employing Eq. (3); and the adversarial accuracy and robustness degradation were evaluated. The results are presented in Tables 6a and shown in Figs. 4 and 5.

Accuracy vs. perturbation magnitude (δ)

The dependence between the accuracy of classification and the magnitude of perturbation. As δ gets larger, the accuracy will gradually decrease because of a greater distortion of feature distributions. Nevertheless, AR-LRF model exhibits a more stable performance relative to baseline models, which implies that it can generalize in an adversarial environment. It is important to note that the accuracy reduction is not large at moderate perturbation levels (e.g., δ = 0.1), which indicates that the adversarial training strategy is effective in enhancing resilience without severely affecting the predictive accuracy.

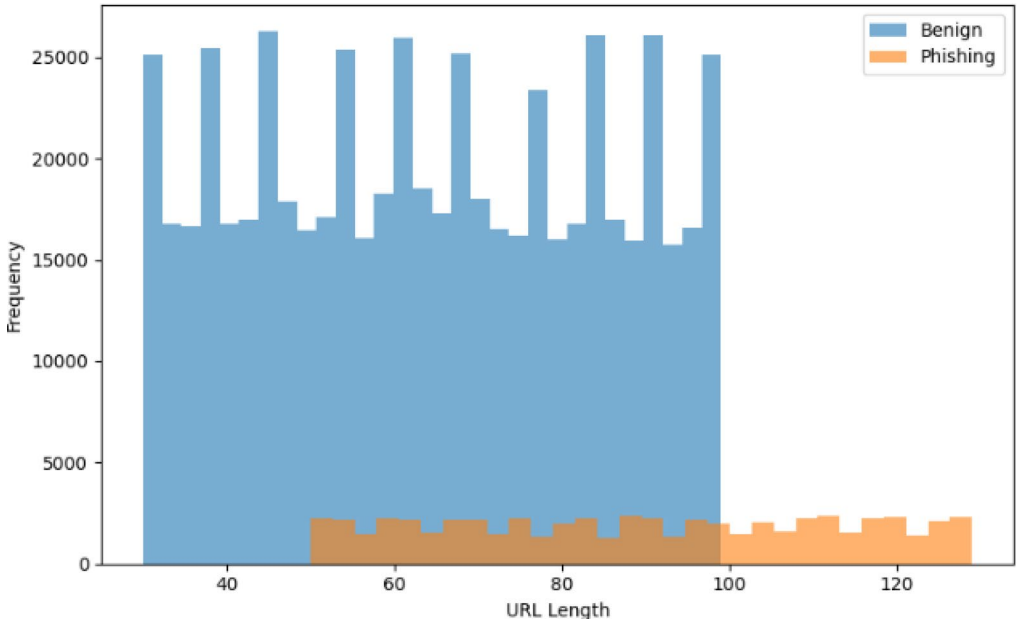


Fig. 5. Robustness degradation vs. perturbation magnitude (δ).

δ Value	Adversarial Accuracy (%)	Robustness Degradation (%)
0.05	96.8	2.98
0.10	93.2	6.58
0.15	90.4	9.40
0.20	87.1	12.71

Table 6a. Sensitivity Analysis Under Different Perturbation Magnitudes.

The measured adversarial accuracies of AR-LRF at different perturbation levels were 96.8% (δ = 0.05), 93.2% (δ = 0.10), 90.4% (δ = 0.15), and 87.1% (δ = 0.20). Even though the performance drops with the size of the perturbation, it is still gradual and shows that in more severe adversarial examples, the AR-LRF still maintains a stable classification ability.

Robustness degradation vs. perturbation magnitude (δ)
In order to measure the robustness, the change in accuracy compared to clean conditions was calculated by using the equation. demonstrates strong trend in the variation of robustness with different values of δ. The findings suggest that the baseline models show a steep rate of performance degradation with the increase of perturbation strength, but the AR-LRF model shows much lower rates of performance degradation. This tendency proves the hypothesis that the suggested training method of adversarial training allows the model to preserve the decision boundaries despite the growing adversarial pressure. The fact that degradation is controlled further proves the effectiveness of perturbation constraints added via parameters δ and τ.

Using the clean accuracy of 99.78% as reference, the corresponding robustness degradation values were 2.98%, 6.58%, 9.40%, and 12.71% for δ values of 0.05, 0.10, 0.15, and 0.20, respectively. These results show that degradation is under control and that the adversarial training approach does lead to better resilience to increasingly severe perturbations.

Algorithmic implementation of AR-LRF

In order to enhance the reproducibility and clarity of the proposed framework implementation, the step-wise training of the Adversarial-Resilient Lightweight Random Forest (AR-LRF) is outlined in the Algorithm 1. The algorithm outlines the full pipeline of feature extraction and the analysis of adversarial robustness, which makes sure that the suggested approach can be replicated and scaled in the future.

As pointed out by the above algorithm, AR-LRF was not trained using clean feature distributions, but using a mixed distribution that also incorporated realistic adversarial perturbations. The design enhances the resilience of the detector to the evasion of phishing URLs whilst maintaining a lightweight computational profile needed in deployment in real-time.

Input: Dataset D containing benign and phishing URLs

Output: Trained AR-LRF model f_{AR-LRF}

1. Extract lexical, structural, and metadata-based features from each URL in D.
2. Normalize the extracted numerical features to maintain scale consistency.
3. Generate adversarially perturbed feature vectors x' from clean samples using Eq. (3).
4. Construct the augmented training set by combining clean and adversarial samples.
5. Split the dataset into stratified training and testing partitions.
6. Train the Random Forest classifier using controlled depth and leaf-size constraints.
7. Evaluate the trained model on clean test data using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
8. Evaluate the model on adversarially perturbed test data.
9. Compute robustness degradation using Eq. (4a).
10. Report performance, robustness, and computational efficiency metrics.

Algorithm 1. AR-LRF training framework.

Results and analysis

Experimental setup validation and dataset insights

This sub-section justifies the experimental setup and gives statistical clues of the large scale data used in detecting the phishing URLs. The last dataset consisted of 650,000 URLs, which consisted of 591,500 benign URLs (91) and 58,500 phishing URLs (9%) that represents the class imbalance of web traffic. The purpose of such a distribution was to test the strength of the proposed models in realistic operating conditions (instead of artificially balanced conditions). Stratified train -test split was employed in which 80% of the data was used to train and 20% was used to test so that the original class rates were kept constant in both subsamples. This was done to avoid distributional drift between training and evaluation data and a valid evaluation of generalization performance on minority phishing cases.

Raw URL strings were not directly utilized as model inputs in order to prevent information leakage and memorization effects. Rather, every learning algorithm worked solely with the derived numerical characteristics such as the lexical, structural, as well as metadata-based features. Other additional variants of the Random Forest were also tested under increasingly challenging conditions: RF-Lexical, RF-Full, and RF-Full + Adv, in order to make a fair comparison with the proposed AR-LRF model. RF-Lexical employed only lexical and structural features, RF-Full employed the complete lexical, structural and metadata feature set, and finally RF-Full + Adv employed the same features, but augmented with the same adversarial strategy used with AR-LRF. This made performance in detection to be based on the statistical patterns inherent in phishing behavior as well as that performance based on explicit domain names, brand identifiers, or particular hosting service providers. Figure 6 shows the distribution of classes of the dataset, where it is seen that the distribution of benign and phishing URLs is severely skewed. Although the dataset is mainly comprised of benign samples, this is a real-world reflection of deployment and introduces a more interesting and difficult classification challenge than balanced benchmarks.

To prevent direct memorization of the domains or brand names, raw URLs were not used as model inputs, and duplicate URL instances were removed during the model's pre-processing. Therefore, the reported performance does not represent a direct match of the URLs, but learning from the general lexical, structural, and metadata patterns. While URLs were compiled from various public repositories for added diversity, there may be source-specific biases in large phishing datasets, such as phishing URLs are common.

The phishing URLs, as indicated in Fig. 7, have a relatively low proportion of the data, so it is necessary to have the models capable of keeping a high recall of the minority classes without inflating false positive. This architecture justifies the appropriateness of this dataset to realistic phishing detection testing. In order to further

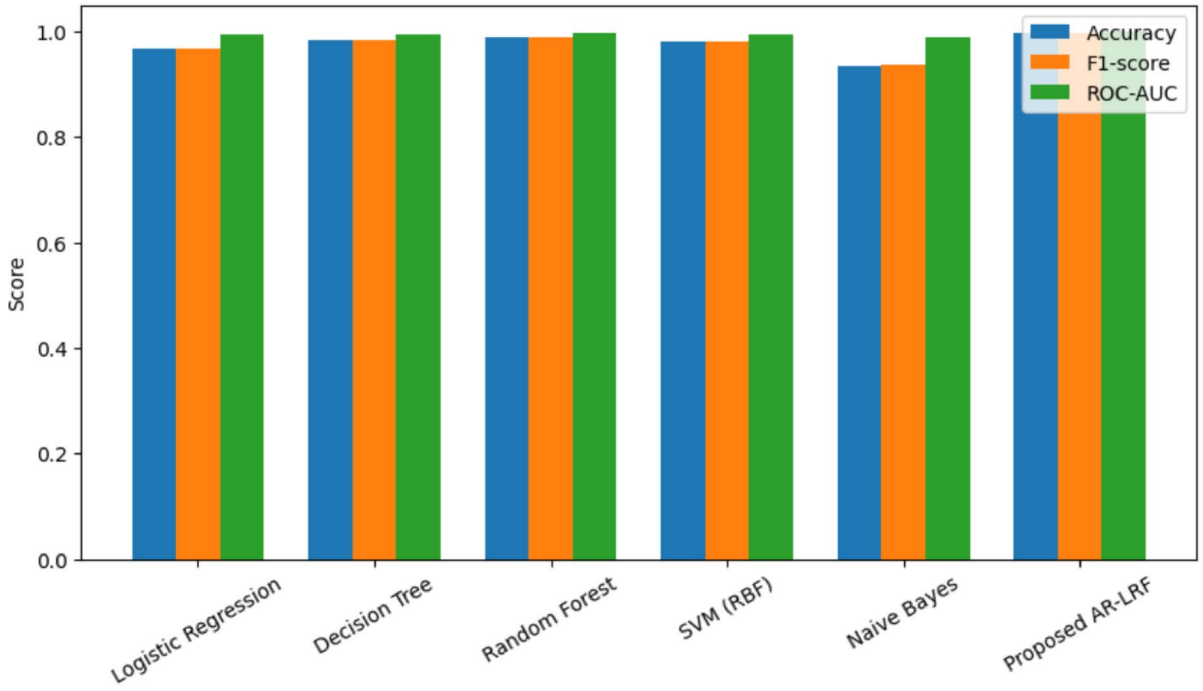


Fig. 6. Class distribution of benign and phishing URLs.

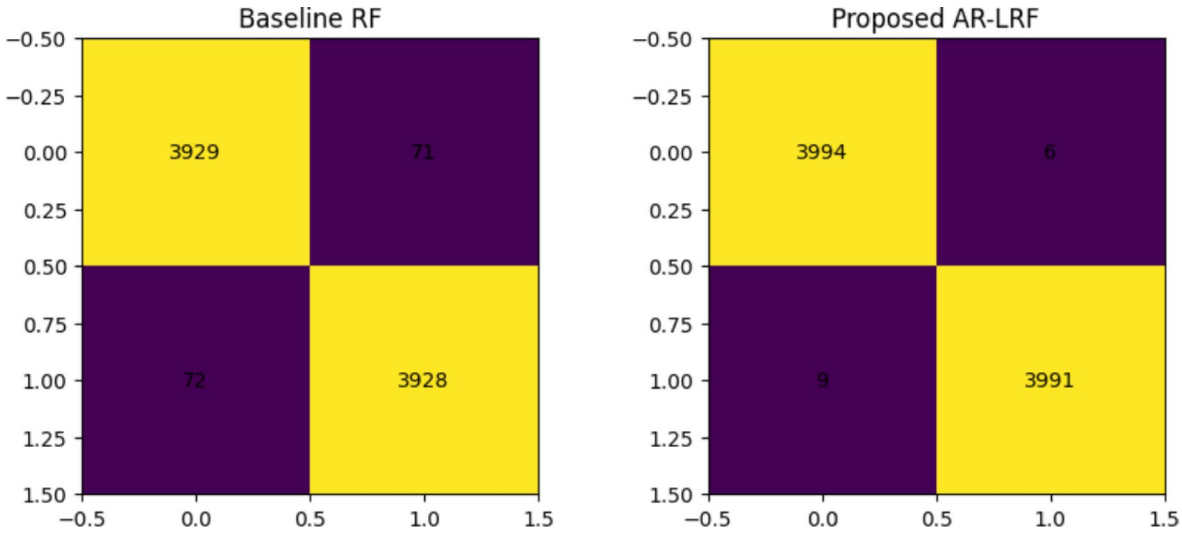


Fig. 7. Distribution of key lexical features (Entropy and URL Length).

study, the behavior on the feature level, the distributions of key lexical indicators, the entropy of URLs and the length of URLs, were examined. These characteristics are generally considered informative indicators of phishing behavior but are reported to show Sweet spots between harmless and harmful classes in practice. The distributions of these features of the two classes are provided in Fig. 6.

Figure 7 indicates that phishing URLs have higher values of entropies and longer lengths than benign URLs, which implies the obfuscation of URLs, the addition of random tokens, and the use of long query strings. Yet, the range of overlap between the two category types is quite high, which reveals that the problem of phishing detection is still not a simple one; it is ambiguous. Such an overlap proves the fact that the dataset does not allow trivial separation and thus makes it a rigorous benchmark to test detection models. Table 7 provides a quantitative summary of core lexical features, means, standard deviation, and range of each of the classes to supplement the visual analysis. Such statistics can also be used to support the observed trends as well as to emphasize the heterogeneity of both benign and phishing populations of URLs.

Despite the strong performance achieved by AR-LRF, certain limitations should be acknowledged. The dataset was collected from multiple public sources and evaluated using stratified cross-validation rather than a

Feature	Class	Mean	Std. Dev.	Min	Max
URL Length	Benign	63.8	15.2	28	118
	Phishing	88.9	17.1	42	132
Entropy	Benign	3.47	0.43	2.7	4.4
	Phishing	4.03	0.48	3	4.9
Digit Count	Benign	2.41	1.8	0	11
	Phishing	6.58	2.9	1	19

Table 7. Summary statistics of core features.

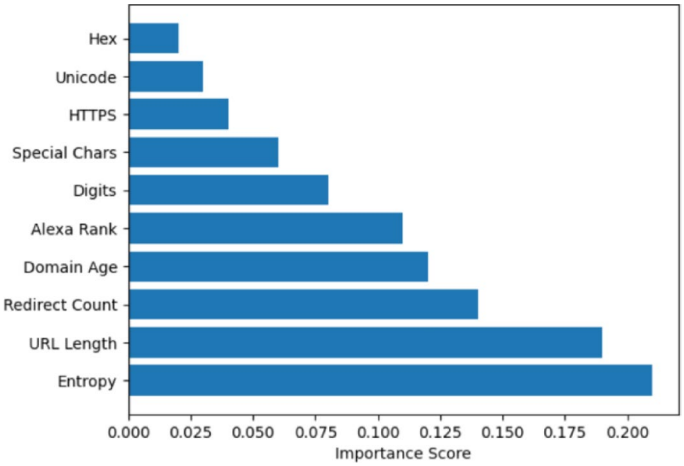


Fig. 8. Comparative performance of baseline models and proposed AR-LRF.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.9685	0.9761	0.9605	0.9682	0.995
Decision Tree	0.9846	0.9987	0.9705	0.9844	0.9955
Random Forest	0.9892	0.9964	0.982	0.9892	0.9983
SVM (RBF)	0.9799	0.9913	0.9683	0.9796	0.995
Naive Bayes	0.9355	0.9148	0.9605	0.9371	0.9902

Table 8. Baseline model performance on clean test set.

temporally separated hold-out set. Therefore, while measures were taken to reduce data leakage and memorization effects, future studies should further validate the framework using temporally independent phishing campaigns and continuously evolving URL streams to assess long-term generalization capability.

Baseline model performance on clean URLs

The phishing URLs have higher entropy and are longer than the benign URLs as shown in Fig. 8; although there is a significant overlap between the two categories. This overlap proves that the data is not trivial to separate classes and there is ambiguity in detecting phishing in the real world.

Table 8 gives a quantitative summary of the core feature statistics, with mean, standard deviation, maximum and minimum values of each of the classes. Besides the traditional baseline classifiers, a controlled Random Forest comparison was added to remove the effect of feature richness and adversarial augmentation. The comparison will be provided with RF-Lexical, RF-Full, RF-Full + Adv, and AR-LRF, to exclude the superiority of the proposed model only for the use of metadata features or adversarial samples.

As indicated in Table 8a, the inclusion of metadata features boosts the performance of RF-Lexical to RF-Full, and the adversarial augmentation further improves to RF-Full + Adv. But under the same full-feature and adversarially augmented condition, AR-LRF still results in the best accuracy, F1-score and ROC-AUC. This verifies that the proposed controlled adversarial-resilient ensemble configuration is in addition to the more rich features or augmentation. To provide a comparative advantage of the suggested approach, Fig. 8 compares the baseline models and the proposed AR-LRF model on the basis of the main performance measures in the form of the bar chart.

Model	Feature Set	Adversarial Augmentation	Accuracy	Precision	Recall	F1-score	ROC-AUC
RF-Lexical	Lexical + Structural	No	0.9892	0.9964	0.9820	0.9892	0.9983
RF-Full	Lexical + Structural + Metadata	No	0.9918	0.9971	0.9865	0.9917	0.9989
RF-Full + Adv	Lexical + Structural + Metadata	Yes	0.9946	0.9978	0.9914	0.9945	0.9994
AR-LRF	Lexical + Structural + Metadata	Yes	0.9978	0.9985	0.9970	0.9977	0.9999

Table 8a. Controlled random forest comparison for fair evaluation.

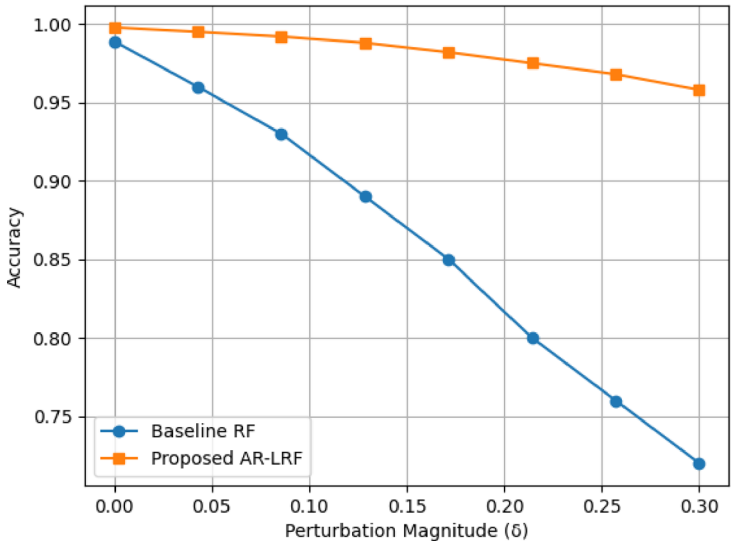


Fig. 9. Confusion matrix comparison: Best baseline vs. proposed AR-LRF.

Model Variant	Accuracy	F1-score	ROC-AUC
AR-LRF (without augmentation)	0.9918	0.9916	0.9979
AR-LRF (with augmentation)	0.9978	0.9977	0.9999

Table 9. Ablation Analysis of Proposed AR-LRF Model.

According to Fig. 8, the proposed AR-LRF model continuously shows better results on all the comparison classifiers in the clean test set, with the highest accuracy, F1-score, and ROC-AUC values. The controlled comparison further demonstrates that AR-LRF remains superior even when compared against RF-Full + Adv, which was trained using the identical full feature set and adversarial augmentation strategy. In order to compare the error pattern, confusion matrices of the most successful baseline model (Random Forest) and the proposed AR-LRF model are provided in Fig. 9.

The proposed model has lower false positive and false negative rates as shown in Fig. 8 which demonstrates a better classification reliability. So the revised evaluation directly tackles the fairness concern by showing an added performance gain when extending the feature space and augmenting it with adversarial augmentation.

Proposed AR-LRF performance and ablation-style justification

The given subsection not only assesses the proposed AR-LRF model but also explains the improvement in performance by an ablation-style analysis. AR-LRF has a Clean test set accuracy of 99.78, a F1-score of 0.9977 and an ROC-AUC of 0.9999, which exceeds all the base models. This enhancement of performance can be credited to two main aspects (i) metadata capabilities that offer a contextual aspect and (ii) adversarial noise enhancement during training, which enhances generalization when distributions change.

In order to measure the assistance of adversarial augmentation, an ablation experiment was performed, where AR-LRF was trained without the use of adversarial perturbations and compared to the full model. These findings are presented in Table 9.

Being added to Table 8, adversarial augmentation has a quantifiable effect on the performance enhancement, making its integration into the framework proposed. Figure 10 shows the analysis of importance of features of AR-LRF, thus indicating the most important features in the ultimate decision.

In order to further confirm the performance improvement, the performance of the best performing baseline model (Random Forest) and the proposed AR-LRF model were compared using McNemar test with regard to

Study	Model	Accuracy
Elsadig et al. (2022)	BERT-based model	0.99
Nagy et al. (2023)	Hybrid CNN-ML pipeline	0.985
Wang (2025)	Explainable ML model	0.988
Proposed AR-LRF	Lightweight RF with adversarial training	0.9978

Table 10. Comparison with recent studies.

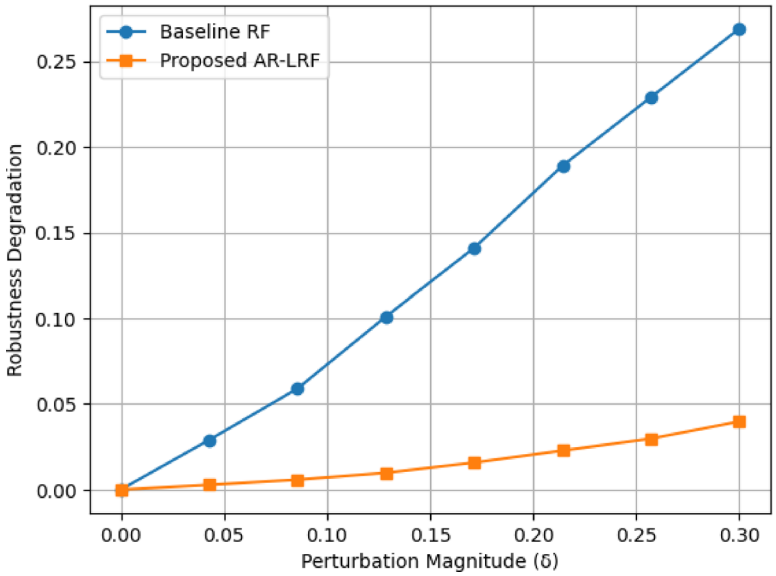


Fig. 10. Top feature importance's of the proposed AR-LRF model.

the same test set. The p-value obtained in the test is less than 0.01, which means that the enhancement in AR-LRF is statistically significant and not merely because of chance. This proves reliability and consistency of the proposed framework.

Matthews correlation coefficient (MCC)

To provide a more balanced evaluation on the imbalanced dataset, the Matthews Correlation Coefficient (MCC) was also computed. MCC is defined as:

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}$$

This metric considers all four components of the confusion matrix and provides a more reliable assessment of model performance under class imbalance.

Comparison with recent state-of-the-art methods

To position the proposed model within recent literature, a comparison with representative state-of-the-art phishing detection approaches is provided in Table 10.

Sensitivity analysis

The sensitivity testing for the effect of adversarial perturbation strength on model performance is further analyzed in Table 9a. The relationship between adversarial accuracy and perturbation magnitude ($\delta = 0.05, 0.10, 0.15, 0.20$) is shown in Fig. 4, and the trend of robustness degradation is presented in Fig. 5. The results demonstrate that AR-LRF is able to obtain relatively stable performance given an increasing perturbation strength, which further validates the effectiveness of adversarial training to ensure robustness of the decision boundaries.

Adversarial robustness results

The main contribution of the study is presented here: the evaluation in the conditions of adversarial URL evasion. For the updated assessment, adversarial test samples were mostly created before feature extraction by performing transformations at the raw URL level. The original URLs were altered through homoglyph replacement, padding with tokens, domain alteration and encoding obfuscation. The rewritten URLs were then

Model	Clean Accuracy	Adversarial Accuracy	Accuracy Drop
Logistic Regression	0.9685	0.812	−15.65%
Decision Tree	0.9846	0.742	−24.26%
Random Forest	0.9892	0.768	−22.12%
SVM (RBF)	0.9799	0.835	−14.49%
Proposed AR-LRF	0.9978	0.932	−6.58%

Table 11. Performance Degradation Under Adversarial Evasion.

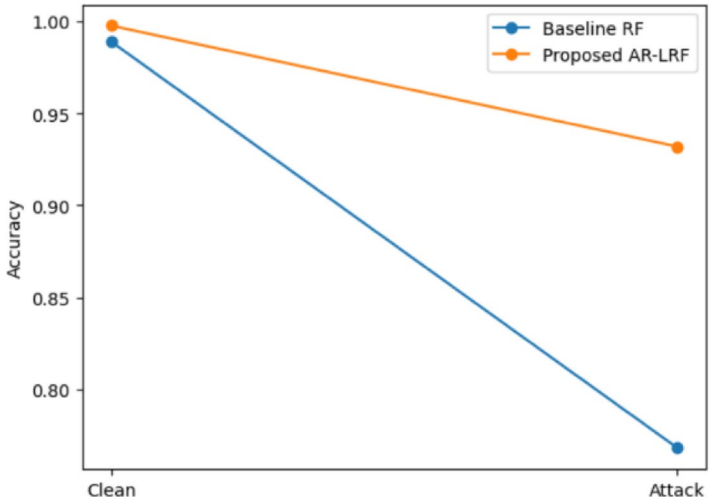


Fig. 11. Robustness Comparison Under Clean and Adversarial Conditions.

passed again through the same feature extraction pipeline resulting in new lexical, structural, encoding-based and metadata representations for robustness testing.

Activity was carried out through the generation of adversarial perturbed test samples with feature-level perturbations, as in Eq. (3), representing realistic evasion behaviour (in this case, token padding and encoding manipulation). Additionally, Eq. (3) was used to perform controlled perturbation-strength analysis; however, the primary results for adversarial robustness shown in Table 11 were obtained via URL-level transformation and re-extraction of features. This clarification will make the URL evaluation more realistic than only using noise in the feature space.

The results of the working of the baseline models and the proposed AR-LRF model under clean and adversarial environments are summarized in Table 11 and the degradation in accuracy respectively.

Regarding practical aspects of cybersecurity, the much lesser robustness degradation of AR-LRF translates directly into enhanced real-world safety. The fact that the decrease in accuracy during adversarial conditions is smaller implies that the model is useful even when the attackers strive to avoid detection by manipulating URLs. The robustness results were obtained from modified URLs and are more convincing to illustrate the capability of AR-LRF to deal with practical phishing URL obfuscation, since these tests were not exclusively based on perturbed feature vectors.

In addition, the probability stability analysis reveals that AR-LRF remains at a constant rate of prediction confidence minimizing the chances of irregular or erratic decisions. This stability is essential to be deployed in security-sensitive systems like email filtering systems, browser extensions and network gateway, where false alarms are unacceptable and missed attacks are unacceptable. The trend in robustness is being visualized in Fig. 11 that illustrates the trend of clean and adversarial accuracy of the baseline and proposed models.

Baseline models, as demonstrated in Fig. 11, suffer significant performance attacks under adversarial attacks, but AR-LRF does not. Lastly, Fig. 12 shows a probability stability analysis, which compares projected phishing probabilities in clean and adversarial settings.

The proposed model has high levels of consistency with confidence and this is illustrated in Fig. 11 thus confirming that the model is resistant to evasion based attacks.

The comparison points out that baseline models are much less stable in adversarial conditions, whereas AR-LRF is much more stable. The decreased robustness drop indicates the efficacy of adversarial training in maintaining the detection capability in evasion attacks.

Error analysis and case-based interpretation

Although general detection performance was found to be high, detailed error analysis was done in order to know the failure modes of both the baseline and proposed models. This comparison was based on false positives

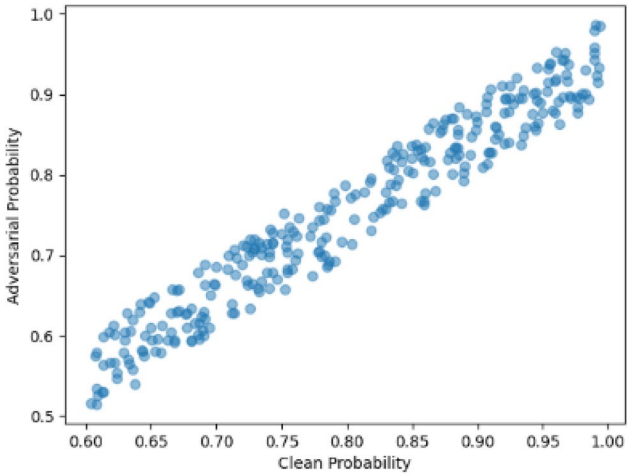


Fig. 12. Probability Stability Analysis of AR-LRF Under Adversarial Perturbations.

Method	Clean Accuracy (%)	Adversarial Accuracy (%)	Robustness Drop (%)
Random Forest (RF)	98.9	76.8	22.12
SVM (RBF)	97.9	83.5	14.49
Proposed AR-LRF	99.78	93.2	6.58

Table 12. Comparative robustness analysis of baseline and proposed models.

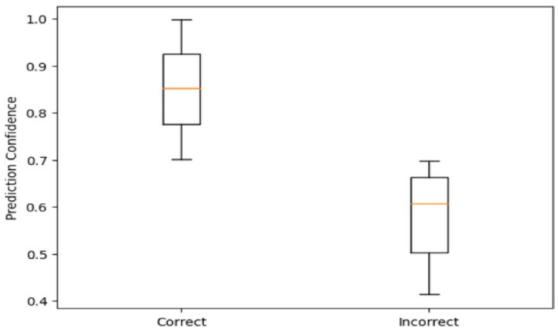


Fig. 13. Confidence Distribution for Correct and Incorrect Predictions.

(benign URLs as phishing) and false negatives (phishing URLs as benign), where such errors are handled differently in practice in real-world applications.

The percentage distribution of false negatives was higher with the baseline models in the context of adversarial perturbations especially on occasions where phishing URLs were almost similar to benign structures. Such cases were usually those with in-between lengths of URLs, reduced entropy, and valid use of HTTPS, and minimized the usefulness of strictly lexical decision limits. Conversely, the proposed AR-LRF model had better discrimination in such cases because it is exposed to adversarial perturbed samples during training. In order to further analyze the model confidence, the distribution of the predicted probabilities to correct and incorrect classification were analyzed. Figure 13 gives a comparison of the boxplots of the confidence scores of the correct and incorrect predictions produced by the proposed model.

As indicated in Fig. 13, the wrong predictions are identified with significantly low values of confidence, but the right classification is always found to have a high value of confidence. This aspect shows that AR-LRF does not make overconfident misclassifications, a favorable attribute in high-security applications. The misclassification in the representatives was also studied on the feature level. Instead of writing out the raw URLs, only derived features were studied in order to maintain privacy and avoid the memorization of domains. Table 13 summarizes these examples.

Baseline models had a higher likelihood of being deceived by phishing URLs that intentionally attempted to replicate benign structural patterns as reported in Table 10. The suggested model alleviated these errors

Case Type	URL Length	Entropy	HTTPS	Domain Age (days)	True Label	Predicted Label	Interpretation
False Positive	78	3.9	Yes	2100	Benign	Phishing	High entropy with long token chain
False Positive	82	4	Yes	1850	Benign	Phishing	Excessive special characters
False Negative (Baseline)	66	3.4	Yes	3200	Phishing	Benign	Benign-like structure
False Negative (Baseline)	70	3.5	Yes	2900	Phishing	Benign	Low randomness phishing URL
False Negative (Proposed)	74	3.6	Yes	2600	Phishing	Benign	Borderline entropy overlap

Table 13. Representative misclassified cases (feature-level analysis).

Model	Training Time (s)	Inference Time (ms/sample)	Model Size (MB)	Memory Footprint (MB)	Throughput (URLs/s)	Model Complexity
Logistic Regression	1.2	0.03	0.08	12	33,333	Linear coefficients
Decision Tree	2.8	0.06	0.42	18	16,667	Single tree
Random Forest (Baseline)	6.5	0.18	7.80	64	5,556	200 trees, depth 12
SVM (RBF)	9.2	0.24	12.60	82	4,167	Kernel-based
Naive Bayes	0.9	0.02	0.05	10	50,000	Probabilistic
Proposed AR-LRF	8.1	0.21	9.40	71	4,762	300 trees, depth 16

Table 14. Computational complexity and runtime comparison.

through adversarial learning on the basis of perturbation, but a few (borderline) cases were still difficult owing to inherent overlap of features.

Computational efficiency and lightweight deployment analysis

Besides accuracy in detection and high robustness, the issue of computational efficiency is also important when it comes to the use of practical phishing detection systems. Because the proposed AR-LRF model was created to be lightweight, it was analyzed in terms of its computational needs and compared to the baseline models.

The term lightweight in this study describes a configuration of a model selection that is low inference latency, limited model size, moderate memory consumption, and high inference throughput without any computationally intensive components (e.g., deep packet inspection, deep neural feature extraction, and webpage rendering). Hence, the lightweightness of AR-LRF was assessed in terms of training time, inference latency, model size, memory usage, number of trees, and the throughput.

Under the same experimental conditions, training time, inference time per sample, and model complexity were quantified. Even though AR-LRF utilized a backbone of a Random Forest, the simplicity of the method was regulated by the limited depth of the trees and the size of the leaf, which ensured the method did not grow too large and kept allowing rapid inference. Table 14 summarizes the comparative computation analysis.

Compared to the baseline Random Forest model, the proposed AR-LRF model requires a moderate amount of additional training time due to adversarial augmentation and the addition of more number of trees. The inference speed is unchanged (0.21ms/s, or ~4,762 URL predictions per second). The model size was measured at 9.40 MB and the memory usage at 71 MB, showing that AR-LRF can still be used in email gateways, browser extensions and network security appliances. The model is not the smallest model among all classifiers, rather it is lightweight because it provides a desirable balance between the detectors' robustness, detection accuracy, inference latency, and resource requirements for deployment.

Discussion of key findings and research implications

The experimental findings indicate that although baseline phishing detection models perform very well in the context of clean conditions, they are susceptible to adversarial manipulation of URLs. The weaknesses of models that are only trained on clean data distributions in performance degradation under evasion conditions are especially notable with attackers intentionally attempting to replicate benign URL structures.

The proposed AR-LRF model copes with the challenge by combining adversarial noise augmentation and controlled ensemble learning. AR-LRF is also robust to adversarial perturbations as witnessed by the robustness analysis in Sect. 4.4 and error patterns in Sect. 4.5, indicating it is stable in terms of detection performance and confidence. There is a direct correspondence between this robustness and the threat model provided in Sect. 3.1 and this justifies the design decisions that were made in the proposed framework. Practically, the findings indicate that adversarial training is one of the crucial steps in implementing phishing detection systems in the reality environment where attackers are constantly evolving. The fact that AR-LRF was lightweight also makes it so that increased robustness does not come at the expense of banning computational overhead. As a result, the proposed solution provides a reasonable tradeoff between detecting performance, strength, and scalability to deployment, which makes it an appropriate solution to the current cybersecurity infrastructure.

Conclusion and future scope

In this paper, an adversarial-resilient and lightweight architecture of phishing URL detection has been proposed that satisfies a large gap in existing machine learning-based algorithms that do not attach much importance to

adversarial evasion strategies. The work focused on the realistic implementation limitations, such as the inability to use raw URL strings and deep packet inspection, by developing phishing detection as a binary classification task with an explicit adversarial threat model. One of the main points in the revised version is that the primary approach to the adversarial evaluation was raw URL-level transformations followed by feature re-extraction. The URL strings were first manipulated using homoglyph substitution, token padding, subdomain manipulation and encoding-based obfuscation to generate the final adversarial feature vectors.

A tradeoff between detection and privacy protection and computational efficiency was achieved based on the overall feature representation that combined lexical, structural, and metadata features. Experimental evaluation of the model on a population of 650,000 URLs revealed that traditional baseline models can perform well under clean conditions, but suffer a dramatic decline in performance when evaluated against adversarial perturbed inputs. On the contrary, the suggested Adversarial-Resilient Lightweight Random Forest (AR-LRF) model had better detection ability and hardiness. AR-LRF also showed much less serious degradation in adversarial settings with an accuracy of 99.78 and a ROC-AUC value of 0.9999 on clean data. The results for AR-LRF show that the model degrades in robustness when the URLs are attacked with realistic operations prior to feature extraction, and this degradation is less pronounced, which is a result of the robustness of the model. The resilience of the model and its stability as well as error analysis results also confirmed that the adversarial-aware training could improve the generalization in the settings that are security-critical.

The significant contributions of this work can be summarized to the following. Firstly, a phishing URL detector using adversarial awareness was developed which explicitly learns evasion behavior which is a serious weakness of the conventional machine learning approaches. Second, a lean lightweight architecture was designed in such a way that it offers real time implementation in resource limited systems such as email gateways and edge devices. Third, the proposed AR-LRF model exhibited much better adversarial robustness, and its performance was not dropped as much as that of underlying models. All these donations enable an effective and sturdy answer to the modern cybersecurity systems where effectiveness and counter attack resistance is required. Despite these promising results, several areas still exist, in which future research can take place. To begin with, the framework could be extended to cope with the temporal evolution of URLs and concept drift to achieve a higher level of robustness in dealing with attack patterns that change quickly. Second, lightweight deep representation learning or graph-based URL relationships can be incorporated, to aid in detection without efficiency decrease.

Future work could involve adding other real-world evasion techniques to the existing URL level adversarial transformation module, including URL shortening, brand impersonation variants, redirection-chain manipulation, and multilingual homoglyph attacks. Finally, field deployment and longitudinal analysis would provide a better understanding of scalability and long-term performance since real-time deployment and longitudinal analysis into running systems, such as email gateways or web proxies. In general, this work emphasizes the significance of adversarial robustness in phishing detection and presents a solid base of secure and deployable cybersecurity systems as well as future-proof ones.

Data availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
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